UNIT IV : Heat Exchangers

Topics: types, parallel / counter / cross flow, overall U, fouling, LMTD method, ε–NTU method, solved problems

4.1 Introduction & Classification

A heat exchanger is a device that transfers heat between two (or more) fluids at different temperatures, usually separated by a solid wall, without mixing (e.g. radiators, condensers, boilers, evaporators, oil coolers, intercoolers).

Basis	Classes
Heat-transfer process	Direct-contact (cooling towers, jet condenser) | Indirect: recuperator (wall separates), regenerator (storage matrix, alternately hot/cold)
Flow arrangement	Parallel flow | Counter flow | Cross flow (mixed / unmixed) | multi-pass, mixed
Construction	Double pipe, shell-and-tube, plate, plate-fin, tube-fin (compact)
Phase	Single-phase | Two-phase (evaporator, condenser)
Compactness	Compact if $A/V > 700\ m^2/m^3$ (car radiators ≈ 1100, human lung ≈ 20 000)

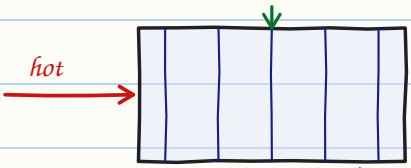
Flow arrangements



PARALLEL FLOW (same direction)



COUNTER FLOW (opposite)



CROSS FLOW

Hot fluid : red Cold fluid : green
Cross-flow : one fluid mixed or unmixed

Common constructions

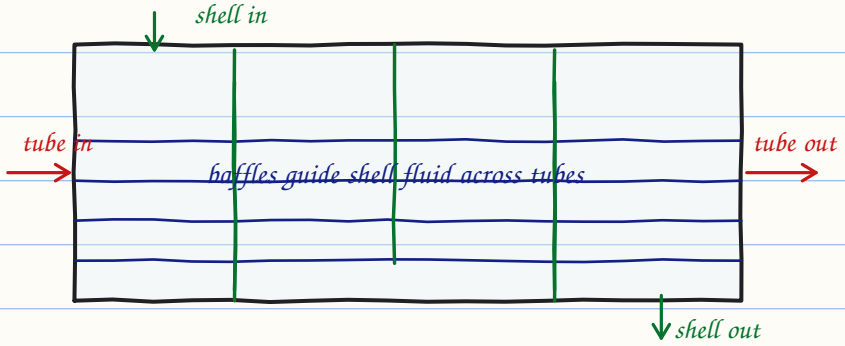
→ Double-pipe (pipe-in-pipe) : simplest; for small duties; true counter/parallel flow; hairpin sections

→ Shell-and-tube : bundle of tubes in a cylindrical shell with baffles. Designated by passes (1-1, 1-2, 2-4). TEMA types: fixed tube sheet, U-tube, floating head. Most common in industry (oil refining, power plants).

→ Plate heat exchanger : corrugated plates with gaskets; high turbulence, easy cleaning, limited to ≈ 25 bar, 150°C

→ Compact (plate-fin, tube-fin) : gas-to-gas and gas-to-liquid (radiators, intercoolers, aircraft)

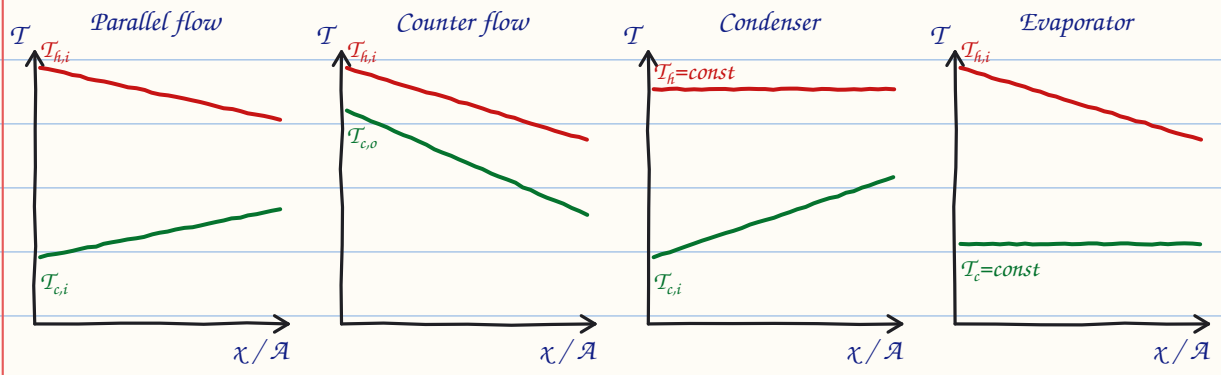
→ Regenerator : rotary wheel (air pre-heater) or fixed matrix; heat stored in matrix between hot and cold periods



1 shell pass – 2 tube passes (1-2 exchanger)

Baffles → cross-flow over tubes, higher h_o, $\Delta P \uparrow$

Temperature Distributions



→ Parallel flow : both fluids enter at the same end; large ΔT at inlet, small at outlet; outlet temperatures can never cross ($T_{c,o} < T_{h,o}$).

→ Counter flow : fluids enter at opposite ends; ΔT fairly uniform; $T_{c,o}$ can exceed $T_{h,o}$; gives highest ΔT_{lm} and best effectiveness for the same A and U.

→ Condenser (hot fluid constant T) and evaporator (cold fluid constant T): arrangement does not matter – $C_{max} \to \infty$, $C_r = 0$.

Overall Heat Transfer Coefficient U

Heat passes through: hot-fluid film → fouling → wall → fouling → cold-fluid film. The resistances add in series. For a plane wall:

$$1/U = 1/h_i + R_{fi} + L/k + R_{fo} + 1/h_o \qquad \text{plane wall}$$

For a tube (inner radius r_i, outer radius r_o, length L) the area must be specified:

$$1/(U_o A_o) = 1/(h_i A_i) + R_{fi}/A_i + \ln(r_o/r_i)/(2\pi k L) + R_{fo}/A_o + 1/(h_o A_o)$$

$$Q = U_o\, A_o\, \Delta T_m = U_i\, A_i\, \Delta T_m \;(U_i A_i = U_o A_o = UA)$$ *tube*

→ When the wall is thin and k high: $1/U \approx 1/h_i + 1/h_o$ → U is dominated by the smaller h (gas side). Remedy: fins on the gas side.

Fluid combination	U (W/m²K)
Water – water	850 – 1700
Water – oil	100 – 350
Steam condenser (water cooled)	1000 – 6000
Steam – light fuel oil	200 – 400
Ammonia condenser	800 – 1400
Air – water (finned)	25 – 60
Gas – gas	10 – 40

Solved Example 0 – Overall U of a double-pipe exchanger

Inner tube D_i = 25 mm, D_o = 30 mm, steel k = 50 W/mK; h_i = 1500, h_o = 800 W/m²K; fouling R_{fi} = 0.0002, R_{fo} = 0.0001 m²K/W. Find U_o and the length needed for Q = 63 kW if ΔT_{lm} = 83.9°C.

$1/U_o = (r_o/r_i)(1/h_i + R_{fi}) + r_o \ln(r_o/r_i)/k + R_{fo} + 1/h_o$

$= 1.2(0.000667 + 0.0002) + 0.015(0.1823)/50 + 0.0001 + 0.00125 = 0.00104 +$

$0.0000547 + 0.0001 + 0.00125 = 0.002445 \rightarrow U_o = 409$ W/m²K

$A_o = Q/(U_o \Delta T_{lm}) = 63\,000/(409 \times 83.9) = 1.84$ m²; $L = A_o/(\pi D_o) = 1.84/(\pi \times 0.03) =$ 19.5 m (i.e. several hairpin sections)

Fouling in Heat Exchangers

Fouling = deposit of unwanted material (scale, sludge, corrosion products, biological growth) on the heat-transfer surface. It adds a thermal resistance R_f (m²K/W) and also restricts the flow (higher ΔP).

$$1/U_{dirty} = 1/U_{clean} + R_{fi} + R_{fo}\,,\; R_f = (1/U_d) - (1/U_c)$$

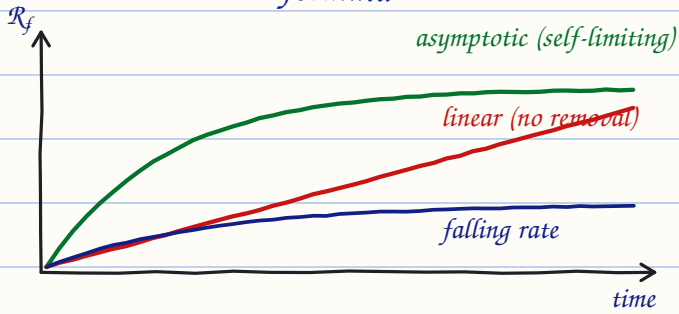
Fluid	R_f (m^2K/W)
Sea water / treated boiler feed water (< 50°C)	0.0001
Sea water (> 50°C)	0.0002
River / city water	0.0002 – 0.0004
Fuel oil	0.0009
Steam (oil-free)	0.0001
Refrigerant liquid	0.0002

→ Types : precipitation (scaling of hard-water salts), particulate, corrosion, biological (algae, bacteria), chemical reaction (polymerisation, coking), freezing

→ Effects : lower U, bigger area needed, higher pumping power, shutdowns, over-design (10–30 %)

→ Control : higher velocity (> 1 m/s), treated water, filters, anti-scalants, tube cleaning (brushes, ball cleaning), choose smooth materials, avoid wall temperature hot-spots, keep dead zones small

→ Cleanliness factor $CF = U_d/U_c$ (0.8 – 0.95 typical); fouling factor adds to resistance in series as in the U formula



Induction → growth → saturation

Net deposition = deposit rate – removal rate (shear)

Higher velocity → higher removal

LMTD Method

Basis : $Q = U A \Delta T_m$, where ΔT_m is a suitable mean of the varying temperature difference between hot and cold streams. Assumptions: steady flow; constant U and c_p; no heat loss to surroundings; no axial conduction; no phase change except in condenser/evaporator; each fluid has uniform properties across section.

Derivation (parallel flow)

Element dA: $dQ = -C_h\,dT_h = C_c\,dT_c$ ($C = \dot{m}\,c_p$, heat-capacity rate), $dQ = U\,\Delta T\,dA$ with $\Delta T = T_h - T_c$:

$$d(\Delta T) = dT_h - dT_c = -dQ\,(1/C_h + 1/C_c) = -U\,\Delta T\,dA\,(1/C_h + 1/C_c)$$

Integrate from inlet (1) to outlet (2): $\ln(\Delta T_2/\Delta T_1) = -UA\,(1/C_h + 1/C_c)$. But $Q = C_h(T_{h1} - T_{h2}) = C_c(T_{c2} - T_{c1})$ so $1/C_h + 1/C_c = [(T_{h1}-T_{h2}) + (T_{c2}-T_{c1})]/Q = (\Delta T_1 - \Delta T_2)/Q$. Hence:

$$Q = U\,A\,\Delta T_{lm}$$

$$\Delta T_{lm} = (\Delta T_1 - \Delta T_2) / \ln(\Delta T_1 / \Delta T_2) \qquad \text{LMTD}$$

Arrangement	ΔT_1 (end 1)	ΔT_2 (end 2)
Parallel flow	$T_{h,i} - T_{c,i}$	$T_{h,o} - T_{c,o}$
Counter flow	$T_{h,i} - T_{c,o}$	$T_{h,o} - T_{c,i}$

→ If $\Delta T_1 = \Delta T_2$, $\Delta T_{lm} = \Delta T_1$ (limit). If $\Delta T_1/\Delta T_2 < 1.7$, the arithmetic mean $(\Delta T_1+\Delta T_2)/2$ is within 2 % of LMTD.

→ Counter-flow ΔT_{lm} > parallel-flow ΔT_{lm} for the same terminal temperatures → smaller area. Condenser/evaporator: same ΔT_{lm} for either arrangement.

→ Counter-flow, $C_h = C_c$: both temperature curves are parallel, ΔT is constant = $\Delta T_1 = \Delta T_2$ and $\Delta T_{lm} = \Delta T_1$.

→ Heat capacity rate $C = \dot{m}\,c_p$ (W/K) : the fluid with smaller C undergoes the larger temperature change. Also: $C_h/C_c = (T_{c,o} - T_{c,i})/(T_{h,i} - T_{h,o})$.

Cross-flow and multi-pass exchangers

Use the counter-flow LMTD multiplied by a correction factor F ($0 < F \le 1$) read from charts for given P and R:

$$Q = U\,A\,F\,\Delta T_{lm,CF}$$

$$P = (t_2 - t_1)/(T_1 - t_1)\,,\quad R = (T_1 - T_2)/(t_2 - t_1) = C_c/C_h$$

For 1 shell pass, 2 (or 2n) tube passes the analytical value is:

$$F = [\sqrt{(R^2+1)}/(R-1)]\,\ln[(1-P)/(1-PR)] \,/\, \ln\{[2 - P(R+1-\sqrt{(R^2+1)})] / [2 - P(R+1+\sqrt{(R^2+1)})]\}$$

→ $F = 1$ for a pure counter-flow, $F = 1$ when one fluid isothermal ($R \to 0$ or ∞). Good design keeps $F > 0.75$; if $F < 0.75$ use more shells in series.

Solved Example 1 – Counter-flow vs parallel flow (area)

Oil ($\dot{m}$ = 0.5 kg/s, c_p = 2100) is cooled from 150°C to 90°C by water ($\dot{m}$ = 0.8 kg/s, c_p = 4180) entering at 25°C. U = 300 W/m²K. Find the area for counter and parallel flow.

Q = 0.5(2100)(60) = 63 kW; water rise = 63 000/(0.8 × 4180) = 18.8°C → $T_{c,o}$ = 43.8°C

Counter : ΔT_1 = 150 - 43.8 = 106.2; ΔT_2 = 90 - 25 = 65; ΔT_{lm} = 41.2/ln(1.634) = 83.9°C; A = 63 000/(300 × 83.9) = 2.50 m²

Parallel : ΔT_1 = 150 - 25 = 125; ΔT_2 = 90 - 43.8 = 46.2; ΔT_{lm} = 78.8/ln(2.705) = 79.1°C; A = 63 000/(300 × 79.1) = 2.65 m² (6 % more)

Solved Example 2 – Shell & tube (1-2) with F

Hot water 120°C → 60°C (C_h = 5 kW/K), cold stream 20°C → 50°C, U = 1000 W/m²K. 1 shell – 2 tube passes. Find A.

Q = 5(60) = 300 kW; ΔT_1 = 120 - 50 = 70, ΔT_2 = 60 - 20 = 40 → $\Delta T_{lm,CF}$ = 30/ln(1.75) = 53.6°C

P = 30/100 = 0.3, R = 60/30 = 2.0 → F = (2.236/1) ln(0.7/0.4)/ln(1.771/0.429) = 1.251/1.417 = 0.883

A = $Q/(U F \Delta T_{lm})$ = 300 000/(1000 × 0.883 × 53.6) = 6.34 m²

Cross-check of Example 2 using the NTU method

C_h = 5 kW/K. C_c = $Q/(\Delta t_c)$ = 300/30 = 10 kW/K → C_{min} = 5, C_r = 0.5. ε = $Q/[C_{min}(T_{h,i} - T_{c,i})]$ = 300/(5 × 100) = 0.60

Counter-flow NTU = $[1/(C_r-1)]$ ln$[(\varepsilon-1)/(\varepsilon C_r-1)]$ = (1/-0.5) ln(0.4/0.7) = 1.119 → UA = 1.119 × 5000 = 5596 W/K. The 1-2 shell exchanger needs UA = 1000 × 6.34 = 6340 → ratio 5596/6340 = 0.883 = F (OK). (Shows F physically = loss of performance versus pure counter-flow).

Effectiveness – NTU Method

The LMTD method needs all four terminal temperatures. When only inlet temperatures are known (rating / performance problems) use the ε–NTU method (Kays & London).

Definitions

$$C_h = \dot{m}_h c_{ph}\ ,\ C_c = \dot{m}_c c_{pc}\ ;\ C_{min},\ C_{max}\ ;\ C_r = C_{min}/C_{max}\ (0 \le C_r \le 1)$$

$$Q_{max} = C_{min}\,(T_{h,i} - T_{c,i})\ \varepsilon = Q/Q_{max} \rightarrow Q = \varepsilon\ C_{min}\,(T_{h,i} - T_{c,i})$$

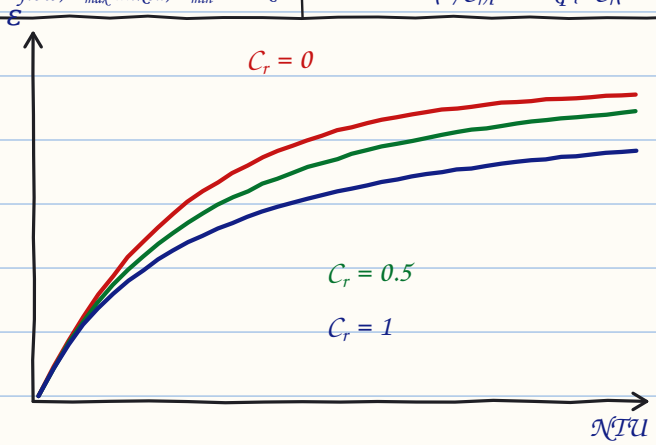
$$NTU = UA / C_{min}$$

→ ε = actual heat transfer / maximum possible ($0 < \varepsilon < 1$). Fluid with C_{min} undergoes the larger temperature change; Q_{max} would need infinite area counter-flow.

→ NTU (number of transfer units) = size of the exchanger in terms of heat-capacity rate: a larger NTU → higher ε with diminishing returns beyond $NTU \approx 3$.

→ $\varepsilon = f(NTU, C_r,$ flow arrangement) :

Flow arrangement	$\varepsilon = f(NTU, C_r)$	$NTU = f(\varepsilon, C_r)$
Parallel flow	$[1 - e^{-NTU(1+C_r)}] / (1+C_r)$	$-\ln[1 - \varepsilon(1+C_r)] / (1+C_r)$
Counter flow	$[1 - e^{-NTU(1-C_r)}] / [1 - C_r e^{-NTU(1-C_r)}]$	$1/(C_r-1) \cdot \ln[(\varepsilon-1)/(\varepsilon C_r-1)]$
Counter, $C_r = 1$	$NTU / (1 + NTU)$	$\varepsilon / (1 - \varepsilon)$
Any, $C_r = 0$ (boiler/condenser)	$1 - e^{-NTU}$	$-\ln(1 - \varepsilon)$
Shell-tube 1 shell pass (2,4.. tube passes)	$2\{1 + C_r + \sqrt{(1+C_r^2)} \cdot (1+e^{-\Gamma})/(1-e^{-\Gamma})\}^{-1}$, $\Gamma = NTU\sqrt{(1+C_r^2)}$	—
Cross flow, both unmixed (approx.)	$1 - \exp\{NTU^{0.22}[e^{-C_r NTU^{0.78}} - 1] / C_r\}$	—
Cross flow, C_{max} mixed, C_{min} unmixed	$(1/C_r)[1 - \exp\{-C_r(1 - e^{-NTU})\}]$	—



Counter-flow exchanger

- ε rises steeply for $NTU < 2$
- little gain beyond $NTU \approx 3$–4
- lower C_r → higher ε
- $\varepsilon = 1$ only if $NTU \rightarrow \infty$, $C_r<1$

Remember: Parallel flow : the maximum possible $\varepsilon = 1/(1 + C_r)$ ($NTU \rightarrow \infty$), e.g. only 0.5 when $C_r = 1$; counter-flow can reach 1. This is why counter-flow is superior.

Derivation of ε for parallel flow

From the LMTD derivation: $\ln(\Delta T_2/\Delta T_1) = -UA(1/C_h + 1/C_c)$. Let $C_{min} = C_h$ (say) so $C_r = C_h/C_c$: $(T_{h,o} - T_{c,o})/(T_{h,i} - T_{c,i}) = \exp[-NTU(1 + C_r)]$.

Energy balance : $T_{c,o} - T_{c,i} = C_r(T_{h,i} - T_{h,o})$ and $T_{h,o} - T_{c,o} = (T_{h,i} - T_{c,i}) - (1 + C_r)(T_{h,i} - T_{h,o})$.

Substituting, with $\varepsilon = (T_{h,i} - T_{h,o})/(T_{h,i} - T_{c,i})$:

$$1 - \varepsilon(1 + C_r) = \exp[-NTU(1+C_r)] \Rightarrow \varepsilon = \{1 - \exp[-NTU(1+C_r)]\} / (1 + C_r)$$ parallel flow

Counter-flow : repeat with $\Delta T_1 = T_{h,i} - T_{c,o}$, $\Delta T_2 = T_{h,o} - T_{c,i} \rightarrow \ln(\Delta T_2/\Delta T_1)$... giving $\varepsilon = (1 - e^{-NTU(1-C_r)}\})/(1 - C_r e^{-NTU(1-C_r)}\})$.

Mixed and unmixed cross-flow

→ Unmixed : fins / channels stop the fluid moving transversely, so temperature varies in the direction normal to the flow. Mixed : fluid free to mix (a bare-tube bank) → uniform temperature across the stream at any section.

→ Both unmixed (plate-fin core) > one mixed > both mixed (for the same NTU); the stream with C_{max} mixed / C_{min} unmixed is better than the opposite.

Procedure – NTU method (rating problem)

1. Compute C_h, $C_c \rightarrow C_{min}$, C_{max}, C_r. 2. $NTU = UA/C_{min}$. 3. Read ε from the formula/chart for the arrangement.

4. $Q = \varepsilon\, C_{min}(T_{h,i} - T_{c,i})$. 5. $T_{h,o} = T_{h,i} - Q/C_h$; $T_{c,o} = T_{c,i} + Q/C_c$.

Solved Example 3 – Counter-flow, NTU method

Hot oil $C_h = 2400$ W/K at 160°C; cold water $C_c = 4000$ W/K at 30°C; counter-flow with $UA = 4200$ W/K. Find Q and outlet temperatures.

$C_{min} = 2400$, $C_{max} = 4000$, $C_r = 0.6$; $NTU = 4200/2400 = 1.75$

$\varepsilon = [1 - e^{-1.75(0.4)}]/[1 - 0.6\, e^{-1.75(0.4)}] = (1 - 0.4966)/(1 - 0.2979) = 0.717$

$Q = 0.717(2400)(160 - 30) = 223.7$ kW; $T_{h,o} = 160 - 223\,700/2400 = 66.8$°C; $T_{c,o} = 30 + 223\,700/4000 = 85.9$°C

(For parallel flow $\varepsilon = [1 - e^{-1.75(1.6)}]/1.6 = 0.587 \rightarrow Q = 183$ kW, i.e. 18 % less.)

Solved Example 4 – Condenser ($C_r = 0$)

Steam condenses at 100°C on tubes carrying water ($\dot{m}$ = 2 kg/s, c_p = 4180 → C = 8360 W/K) entering at 20°C. U = 2500 W/m²K, A = 5 m².

$NTU = 2500(5)/8360 = 1.495$; $\varepsilon = 1 - e^{-1.495} = 0.776$; $Q = 0.776(8360)(100 - 20) =$ 519 kW; $T_{c,o} = 20 + 0.776(80) = 82.1°C$

Solved Example 5 – Design with fouling

A clean exchanger has U_c = 850 W/m²K. After service, fouling resistances R_{fi} = 0.0002 and R_{fo} = 0.0001 m²K/W develop.

$1/U_d = 1/850 + 0.0002 + 0.0001 = 0.001176 + 0.0003 = 0.001476$ → U_d = 677 W/m²K; $CF = 677/850 = 0.80$ → area must be 25 % larger (850/677 = 1.255) to give the same duty.

LMTD vs NTU – when to use which

Given	Find	Method
4 terminal temperatures (or Q and 3 T)	Area A (sizing / design)	LMTD
Inlet temps, flow rates, A, U	Outlet temps, Q (rating)	ε–NTU
Condenser / evaporator	Either	either; $C_r = 0$ simplifies NTU

Heat exchanger selection – design considerations

- Required duty and temperatures; fluid properties; allowable pressure drop; fouling tendency; corrosion → material (SS, Cu-Ni, Ti)
- Size, weight, cost, maintenance (removable bundles for cleaning), leakage/cross-contamination risk, thermal stresses (floating head or U-tube)
- Put the fouling / corrosive / high-pressure fluid in the tubes (cheaper to clean and pressurise), the viscous or condensing fluid in the shell

Compact heat exchangers – key points

- Plate-fin / tube-fin : area density $\beta = A/V$ up to 6000 m²/m³; used for gas-side with low h. Fin efficiency $\eta_o = 1 - (A_f/A)(1 - \eta_f)$ must be included in U.
- Air-conditioning coils, radiators, aircraft oil coolers, cryogenic exchangers (Re small, ΔP small).

Solved Example 6 – Steam condenser (LMTD)

Steam condenses at 120°C; cooling water ($\dot{m}$ = 3 kg/s, c_p = 4180) heats from 20°C to 60°C; U = 2000 W/m²K. Find A.

$Q = 3(4180)(40) = 501.6$ kW; $\Delta T_1 = 120 - 20 = 100$, $\Delta T_2 = 120 - 60 = 60$; $\Delta T_{lm} = 40/\ln(1.667) = 78.3$°C

$A = 501\,600/(2000 \times 78.3) = 3.20$ m²

Solved Example 7 – Cross-flow, both fluids unmixed (NTU)

Hot gas C_h = 1000 W/K at 400°C; air C_c = 2500 W/K at 30°C; UA = 1500 W/K. C_{min} = 1000, C_r = 0.4, NTU = 1.5.

$\varepsilon = 1 - \exp\{ NTU^{0.22}[\, e^{-C_r NTU^{0.78}} - 1 \,]/C_r \} = 1 - \exp\{ 1.093(0.578 - 1)/0.4 \} = 1 - e^{-1.154} = 0.685$

$Q = 0.685(1000)(370) = 253$ kW; $T_{h,o} = 400 - 253 = 147$°C; $T_{c,o} = 30 + 253\,300/2500 = 131$°C

Solved Example 8 – Unknown outlet temperatures

Counter-flow water-to-water: hot 0.5 kg/s at 90°C, cold 0.25 kg/s at 20°C, U = 1000, A = 2 m². (c_p = 4180). $C_h = 2090$, $C_c = 1045 \rightarrow C_{min} = 1045$, $C_r = 0.5$, $NTU = 2000/1045 = 1.914$.

$\varepsilon = (1 - e^{-0.957})/(1 - 0.5\, e^{-0.957}) = 0.616/0.808 = 0.762$; $Q = 0.762(1045)(70) = 55.8$ kW; $T_{h,o} = 90 - 26.7 = 63.3$°C; $T_{c,o} = 20 + 53.4 = 73.4$°C

Regenerators and multi-pass behaviour

- → *Regenerator* : hot gas heats a matrix (period 1), then cold gas takes heat from it (period 2). Used in gas-turbine recuperation, glass furnaces, rotary air pre-heaters (efficiency 70–85 %).
- → In a 1-2 shell-and-tube unit the tube-side fluid passes once with and once against the shell flow; temperature *cross-over* can occur, but a large cross-over reduces F sharply ($F < 0.75$ → add shells).
- → Using several shells in series approaches true counter-flow: ε for N shells increases and F rises towards 1.

Thermal design procedure (flow chart in words)

1. Fix duty Q from energy balance; find missing temperature. 2. Choose type, fluid allocation, tube size & velocity.

3. Compute h_i, h_o (Unit II/III correlations) → U with fouling. 4. ΔT_{lm}, F → area $A = Q/(UF\Delta T_{lm})$.

5. Fix number of tubes & length; check Δp. 6. If Δp or velocity not acceptable, change passes / baffles and repeat.

Comparison of exchanger types

Type	Advantages	Limitations
Double pipe	simple, cheap, true counter-flow	small area, bulky for large duty
Shell & tube	robust, high p,T, large duty	large, costly, shell-side fouling
Plate	compact, high U (3000–7000), easy cleaning	gaskets limit p,T (< 25 bar)
Plate-fin / compact	very high A/V, light weight	clean fluids only, difficult to clean

Pressure drop and pumping power

→ Tube side : $\Delta p = [\, f\, L/D + 4\, n_p \,]\, \rho\, u_m^2/2$; more tube passes → higher h but higher $\Delta p \propto u^2$

→ Shell side : baffles increase h_o but also Δp; baffle spacing ≈ 0.2 – 1 shell diameter, 25 % cut

→ Pumping power = $V\Delta p / \eta_{pump}$. Optimum design balances capital cost (area) with operating cost (pumping).

Quick Revision – formulae

Item	Formula
Overall U (plane wall)	$1/U = 1/h_i + R_{fi} + L/k + R_{fo} + 1/h_o$
LMTD	$(\Delta T_1 - \Delta T_2)/\ln(\Delta T_1/\Delta T_2)$
Duty	$Q = UA\, F\, \Delta T_{lm}$
Effectiveness	$\varepsilon = Q/[C_{min}(T_{h,i} - T_{c,i})]$
NTU	UA/C_{min}
Counter-flow ε	$(1-e^{-NTU(1-C_r)})/(1-C_r e^{-NTU(1-C_r)})$
Fouling	$1/U_d = 1/U_c + R_f$

Quick viva (short answers)

Q1.Why do outlet temperatures cross in counter-flow only? – in counter-flow cold leaves near hot inlet.

Q2.Value of F for condenser? – 1. Value of C_r for evaporator? – 0.

Q3.Which fluid should be on tube side? – the fouling, corrosive or high-pressure one.

Important Questions (exam oriented)

1. Classify heat exchangers. Sketch temperature distributions in parallel and counter-flow exchangers.
2. Derive the LMTD expression for a parallel-flow heat exchanger. State the assumptions.
3. Why is counter-flow better than parallel flow? What is the correction factor F?
4. Define effectiveness and NTU. Derive ε for parallel-flow in terms of NTU and C_r.
5. What is fouling? List its types and effect on overall heat transfer coefficient.
6. Distinguish LMTD and NTU methods with a worked problem each.

